Rheeya Uppaal

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RESEARCH SUMMARY

- PhD candidate developing model editing and **representation-level interventions** for improving language model **safety, truthfulness, and controllability**.
- Studies how failures emerge under **distribution shift, hallucination, and multi-step reasoning**, and how targeted interventions can make model behavior more reliable.

SELECTED PUBLICATIONS

- ***Journey Before Destination: On the importance of Visual Faithfulness in Slow Thinking.*** (EACL 2026, Oral Track)
  Introduces training-free detection and repair of visually unfaithful reasoning steps in VLM chains.
- ***When Safety Fails Before the Answer: Benchmarking Harmful Behavior Detection in Reasoning Chains.*** (Preprint, 2026)
  Builds a sentence-level benchmark for detecting how unsafe behavior emerges inside reasoning traces.
- ***Model Editing as a Robust and Denoised variant of DPO: A Case Study on Toxicity.*** (ICLR 2025) [Talk]
  Connects model editing to preference optimization by showing that safety-aligned edits behave like denoised, sample-efficient DPO updates with greater robustness to noisy preferences.
- ***How Useful is Continued Pre-Training for Generative Unsupervised Domain Adaptation?*** (ACL 2024, RepL4NLP workshop)
  Studies how implicit task learning in continued pretraining improves generative domain adaptation under distribution shift.
- ***Is Fine-tuning Needed? Pre-trained Language Models Are Near Perfect for Out-of-Domain Detection.*** (ACL 2023). [Talk] [Media coverage] Shows pretrained LMs can achieve near-perfect OOD detection under domain shift without fine-tuning.

RESEARCH EXPERIENCE

- **AWS (Amazon) AI Labs:** *Applied Scientist Intern (New York, June-August 2025).* Under Dr Phu Mon Htut.
  Developed methods for process-level evaluation and repair of visually unfaithful reasoning in VLMs, improving reliability of multimodal chains without additional training.
- **Amazon Science:** *Applied Scientist Intern (Seattle, June-August 2024).* Under Dr Markus Dreyer and Prof Mohit Bansal.
  Studied model editing for hallucination reduction in RAG systems, with emphasis on targeted, interpretable interventions over full-model fine-tuning.
- **Goldman Sachs CoreAI:** *Research Engineer (New York City, 2019-2021).* Under Dr Vijay Saraswat.
  Developed information extraction models for high-stakes financial and legal domains, and analyzed arithmetic behavior in BERT-style models for interpretability.
- **UMass Amherst:** *Research Assistant (Amherst, 2018-2020).* Under Prof Andrew McCallum and Prof Madalina Fiterau.
  Researched active learning and domain adaptation for NLP, contributing to work on low-supervision reliability and robust generalization.

SELECTED TALKS

- From Internals to Outcomes: Enhancing Truthfulness and Safety in Language Models. Freestone Grove, 2025.
- Towards Robust Model Editing Techniques. Cohere for AI's Research Connections Community, 2025.
- Model Editing for Aligning Models to Preferred Behaviours. Amazon Science GPS Reading Group, 2024.

AWARDS & LEADERSHIP

- Semi Finalist, *Qualcomm Innovation Fellowship*, 2026.
- *Karamveer Chakra Award* (instituted by iCONGO and the **United Nations**), 2021.
- Research Lead, Wisconsin AI Safety Initiative (2024-26).
- Organizer, workshop on *Trustworthy AI: Validity, Fairness, Explainability, and Uncertainty Assessments*, 2024.
- Mentor: ACM-W, ML-X Marathon, UMass Industry Research Mentorship.

EDUCATION: PhD, Computer Science, UW–Madison | MS, UMass Amherst | BTech, Symbiosis Institute of Technology

TECHNICAL SKILLS: Python | PyTorch, HuggingFace | Model editing, AI Safety, LLM evaluation, Domain adaptation